Countering Vendor Lock-In

Procurement models, calibration, and contract recommendations · Project summary for coauthors · Draft, October 14, 2026

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This document combines the repository README (Part I) and the contract-design recommendations (Part II) of github.com/jaeyk/contract_simulation. File references link to the repository. All results are theoretical or illustrative; nothing here establishes misconduct, intent, vendor profits, or causal effects.

Part I. Project overview, workflow, and results

Game-theoretic models of how procurement terms shape vendor behavior, public spending, and applicant burden. This is a companion to *Profits and Burdens: Artificial Intelligence and the American Contracting State*.

The project asks one question: **when does buying compliance technology become persistent vendor dependence, and weak incentives to reduce burden?** It answers in four layers:

1. a conceptual memo,
2. formal models,
3. simulations and contract-design optimization,
4. a calibration to real state contracts.

Everything here is theory and illustration. No result establishes misconduct, intent, actual vendor profits, or causal effects.

Repository layout

Path	Contents
recommendations.md	Contract-design recommendations and checklist for agencies, the limits of contractual safeguards, and how design priorities differ across legacy, predictive, generative, and agentic systems
manuscript/vendor_exploitation.md	Twelve procurement vulnerabilities, their generative/agentic AI variants, and priority hypotheses
game_theory/README.md	Model 1: competition, dependence, and amendment bargaining
game_theory/extensions.md	Models 2–9, optimal contract design, and calibration method
simulation/	Python solvers, parameter sweeps, design optimizer, calibration, and tests
simulation/calibration/	Hand-verified contract records, clause codes, and coding rules
docs/figures/	Explanatory figures for this README, generated by `simulation/make_figures.py`
simulation/outputs/	Timestamped simulation and calibration runs; earlier runs are preserved
dashboard/	Static browser lab for all nine models; it re-solves equilibria live
`data/`	81 contract PDFs and price catalogs: EPIC FOIA productions and NASPO ValuePoint agreements

Workflow

Step 1. Mechanisms (manuscript)

The memo separates *demand* for compliance technology from the *contractual conditions* that let vendors earn rents. It identifies twelve vulnerabilities:

- lowball bids followed by costly amendments;

- cooperative purchasing without local validation;
- lock-in;
- restrictive licenses;
- payment for verification volume;
- rewarding flags over accuracy;
- continuity-driven extensions;
- policy changes as incumbent revenue;
- reseller layers;
- vendor-shaped specifications;
- weak inspection rights;
- product changes after evaluation.

It also compares how each changes with generative and agentic AI.

Step 2. Formal models (game_theory)

Each model changes one assumption of the baseline, so that its mechanism can be told apart from switching-cost bargaining.

Model	Mechanism	Central result
1	Competition, dependence, Nash-bargained amendments	Competitive bids absorb expected amendment margins; dependence and low quality still waste resources
2	Paid remediation	Paid repair margins weaken prevention unless enforced warranty coverage exceeds $\mu/(\mu+r+f)$
3	Metered workflows	Per-action pay can cause too much work; a fixed fee can cause too little
4	Rewarded flags	The vendor's flagging threshold diverges from the public-value threshold
5	Shared and local validation	Shared testing is underprovided by free riding; local testing is omitted when agencies discount local benefits
6	Human review	The agency cannot commit not to review, which induces low vendor effort and hidden staff labor
7	Post-award substitution	Limited detection makes cheaper, harmful substitution privately attractive
8	Models 1 + 2 combined	Dependence prices uncovered repairs; portability now raises prevention
9	Models 3 + 6 combined	Vendors can hold work at the agency's review threshold so that staff correct their errors

Step 3. Simulation, design, and calibration (simulation)

```
cd simulation
uv sync --locked --cache-dir .cache/uv
.venv/bin/python -m unittest -v                                    # 17 tests
.venv/bin/python run.py                                            # sweeps for Models 1–9
        plus design problems
.venv/bin/python build_visualization.py outputs/<run_dir>          # optional explorer for
        saved results
.venv/bin/python calibration/extract_terms.py                      # only if PDFs or rules
        change; needs pdftotext
.venv/bin/python calibrate.py                                      # calibration report
.venv/bin/python make_figures.py                                   # README figures ->
        docs/figures/
```

- **Equilibrium solvers** (models.py). Continuous problems use analytic boundary checks and bisection on first-order conditions. Model 8's joint choice uses an exact best response for dependence, then a global grid search over effort refined by bisection on the slope of the resulting payoff. Discrete games use backward induction or closed-form mixed equilibria.
- **Sweeps** (config.json). 241 parameter combinations across nine models. Each run writes CSVs, figures, `REPORT.md`, `design.json`, and a manifest with source hashes.
- **Optimal contract design** (design.py). The agency commits to instruments within bounds, such as monitoring, portability, warranty enforcement, prices and penalties. It anticipates the equilibrium they induce, net of quadratic governance costs. The search is a grid with successive zooming, which handles discrete regime changes.
- **Calibration** (calibrate.py) proceeds in four steps:
    1. Hand-verify amendment histories, prices, service-level credit caps, and negotiated deviations, each with a source quote.
    2. Code clauses with patterns that save the matching snippet. Audit the snippets and record manual overrides with reasons.
    3. Map codes to parameters (monitoring m, portability p, penalty exposure F, warranty enforcement w) through editable rules in coding_rules.json.
    4. Match each contract's amendment growth. Report the vendor's bargaining weight θ as an *identified set*: every value consistent with a nonnegative incumbent cost.

Step 4. Interactive exploration (dashboard)

```
simulation/.venv/bin/python dashboard/generate_fixtures.py
node --test dashboard/tests.mjs          # 256 scenarios match Python to 1e-9
simulation/.venv/bin/python -m http.server 8765 --directory dashboard/dist --bind
        127.0.0.1
```

Key outcomes

Theoretical insights, illustrated

How to read these models

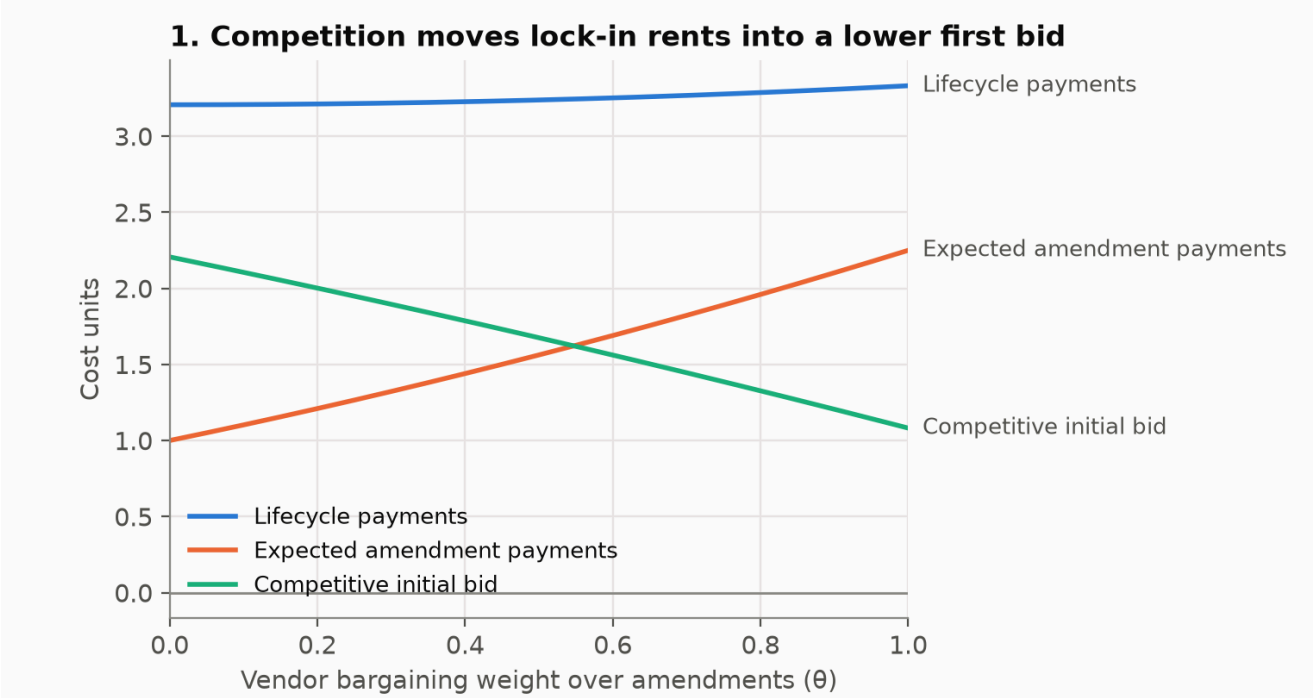
Every model is a game between a **public agency** and a **vendor**.

1. The agency writes the contract: what it pays for, what it checks, and what penalties apply.
2. The vendor then chooses what is most profitable under those terms: how much effort to put into quality, how hard it makes itself to replace, how much billable work to do, or which cases to flag.
3. The agency anticipates those choices.

The **equilibrium** is what both sides actually do. The **planner** benchmark is what someone minimizing society's total real costs would choose; the gap between the two is waste. Money is in abstract "cost units", and parameters are illustrative.

The figures below are solved directly from the models (`simulation/make_figures.py`); Figure 7 plots the contract calibration. **Read the shapes, not the numbers.**

1. A cheap first bid can be paid back through amendments



Model 1, default parameters. Lifecycle rises only by the cost of building dependence; amendment margins are offset in the bid.

The situation. Several vendors compete for a contract. Everyone knows that later changes are likely: new eligibility rules, extra features, more claims. Once the system is running, the agency can't easily switch vendors, so the winner negotiates the price of that later work alone. The horizontal axis, **bargaining power (θ)**, is how much of the gain from that negotiation the incumbent can claim.

What happens.

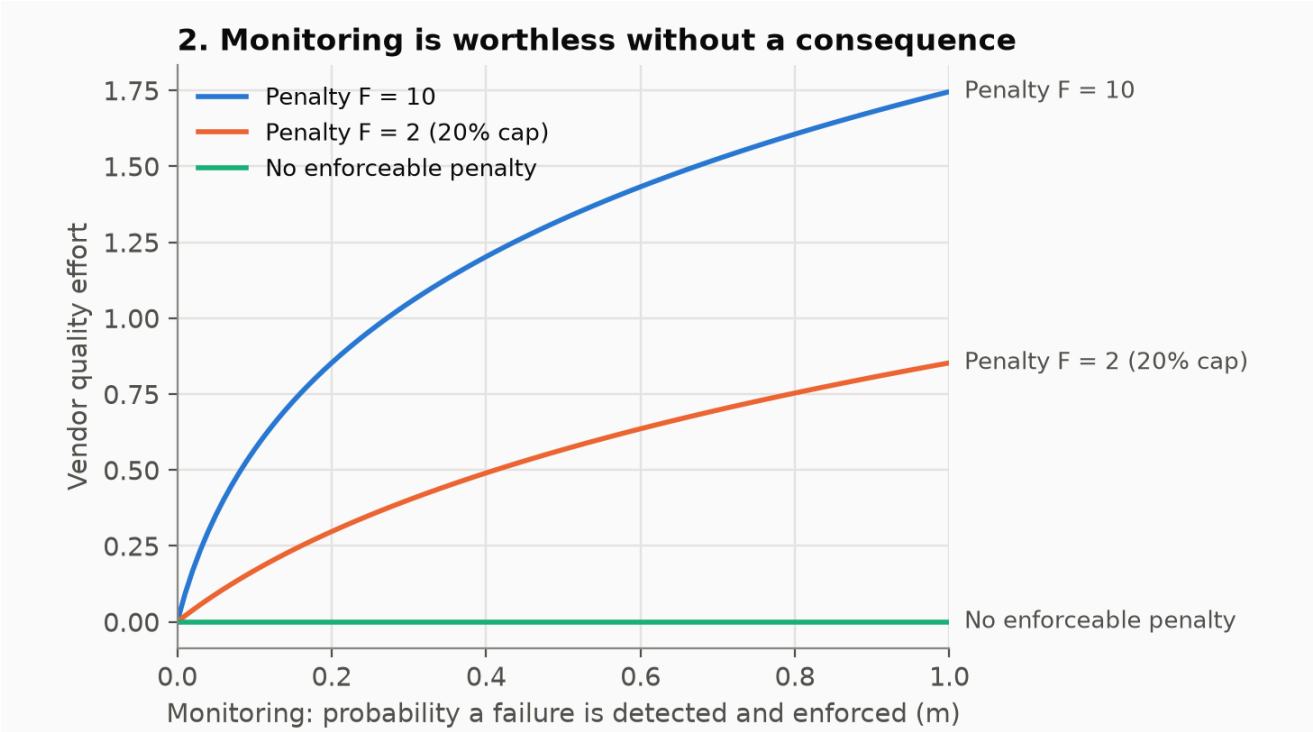
1. The more bargaining power the winner will have, the more profit it expects from future amendments.
2. Because vendors compete for the award, they spend that expected profit up front by bidding lower. It works like a printer sold cheap because the ink is expensive.
3. The initial bid (green) falls, and expected amendment payments (orange) rise by about the same amount.
4. Total payments over the contract's life (blue) barely change. The small rise is real waste: money the vendor spends making itself hard to replace.

Why it matters.

- A low award price is not evidence of a good deal.
- Expensive amendments are not, by themselves, evidence of excess profit.
- Agencies should compare bids on expected **lifecycle** cost, and they should treat lock-in itself as the problem.

In 5 of 25 default scenarios, competition would push the initial bid below zero.

2. Monitoring does nothing unless a failure costs the vendor something



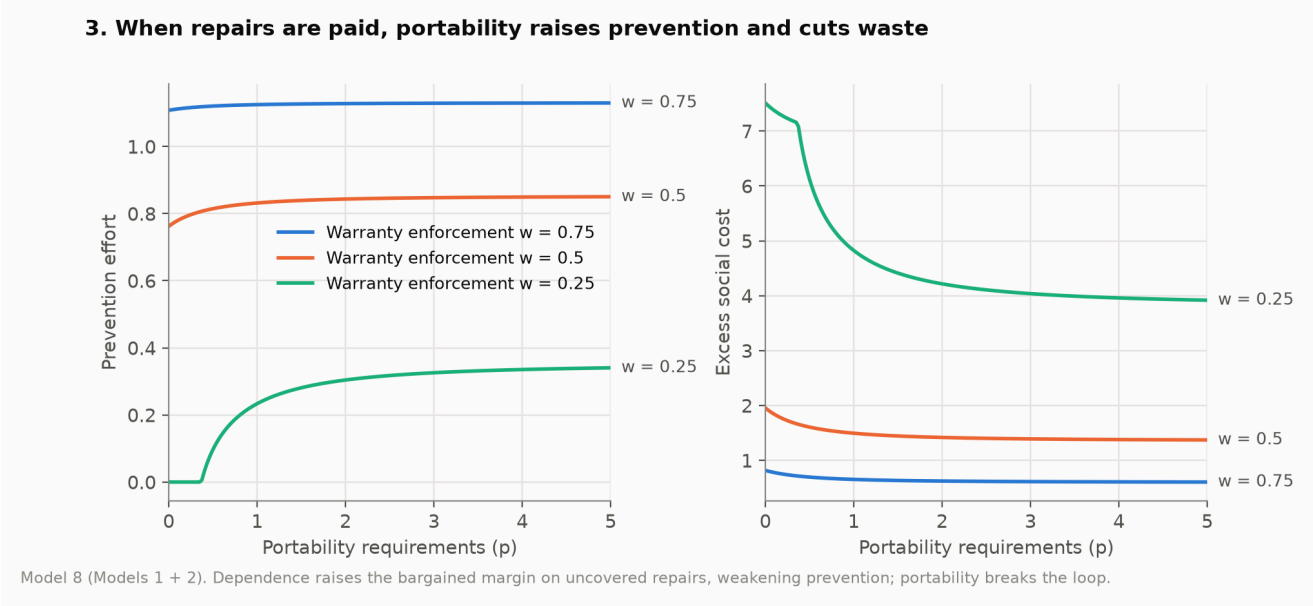
Model 1. Effort solves c·e = m·F·exp(−e): with F = 0 it is zero at every m. Three of four calibrated contracts had no enforceable penalty.

The situation. The vendor decides how much to spend on quality. Failures hurt applicants, but the vendor feels them only if it pays for them. What it expects to pay is the **chance a failure is caught and enforced (m, horizontal axis)** times the **penalty when it is (F)**.

What happens. With a penalty, more monitoring means a failure is more likely to cost the vendor, so it invests more in quality (blue and orange curves). With no enforceable penalty, the expected cost of failure is zero however closely the agency watches, so effort stays at zero (green). It is like a speed camera that never issues tickets.

Why it matters. Write enforceable remedies first, then pay for detection. Three of the four calibrated contracts had no enforceable performance penalty. A penalty at the largest service-level cap found in the contracts (20% of fees, orange) restores substantial effort.

3. Lock-in and defects feed each other, and portability breaks the loop



Model 8 (Models 1 + 2). Dependence raises the bargained margin on uncovered repairs, weakening prevention; portability breaks the loop.

The situation. Some defects are inevitable, but the vendor can prevent many with effort. Two things can happen to a defect:

- **Covered by the warranty:** the vendor fixes it at its own cost.
- **Not covered:** the agency pays for the repair as an amendment, at a price negotiated with a vendor it can't easily replace.

Warranty enforcement (w) is the share of defects actually treated as warranty repairs. That depends on who decides what counts as a defect. **Portability (p, horizontal axis)** is how easily the agency could hand the system to a different vendor.

> **Is portability the same as open source?** Not quite. Open source is one route to portability, but neither guarantees the other. Portability means another vendor could take over without rebuilding from scratch, which requires:
>
> - **Data:** the agency owns its data and can export it in usable, documented formats.
> - **Code:** the agency owns, or holds a license to, custom code built with public money. That can mean an open-source license, but it can also mean a proprietary product backed by source-code

escrow the agency can actually use.
- **Interfaces:** open, documented connections to other systems, rather than proprietary ones only the incumbent understands.
- **Know-how:** documentation, knowledge transfer, and transition assistance at the end of the contract.

Open-sourcing code helps because any vendor can maintain it, but a system can be open source and still locked in through undocumented configurations, vendor-controlled hosting, or knowledge held only by the incumbent's staff. Conversely, a proprietary product can be fairly portable if data export, open interfaces, and transition help are guaranteed. In the Arizona UI contract, the vendor negotiated **source code** out of the deliverables and barred the state from inspecting the code escrow, which lowered portability even though data ownership stayed with the state.

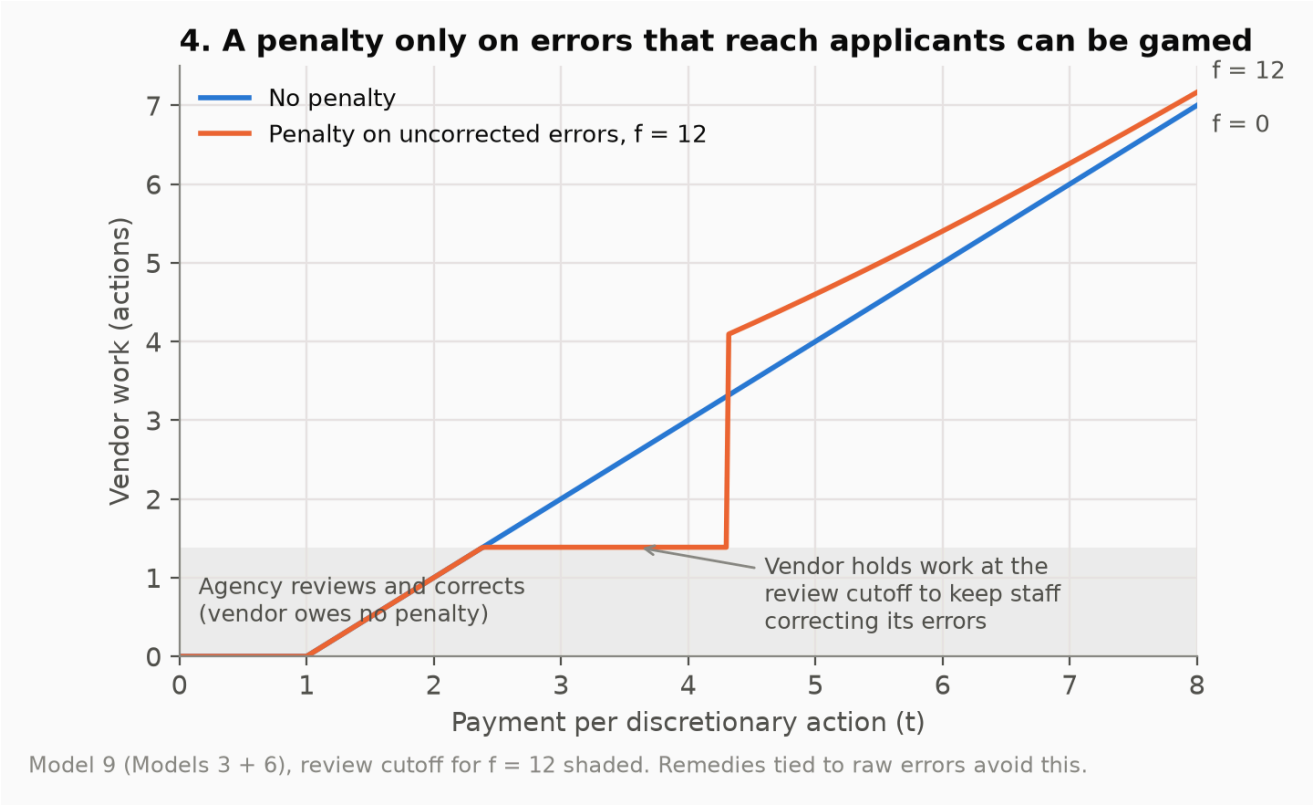
What happens.

1. The harder the vendor is to replace, the more it can charge for each paid repair.
2. The more profitable paid repairs are, the less it pays to prevent defects.
3. More defects make lock-in even more valuable to the vendor, and the loop closes.

Portability cuts the repair margin and restores the incentive to prevent defects (left panel). With weak warranty enforcement (green), prevention is zero until portability passes a threshold, then rises; waste roughly halves, from 7.5 to 3.9 (right panel).

Why it matters. Warranties and portability work as a pair: either one is weaker without the other. Agencies should keep sole authority to classify defects, since that is what makes a warranty real.

4. A penalty only on errors that reach applicants can be gamed



Model 9 (Models 3 + 6), review cutoff for f = 12 shaded. Remedies tied to raw errors avoid this.

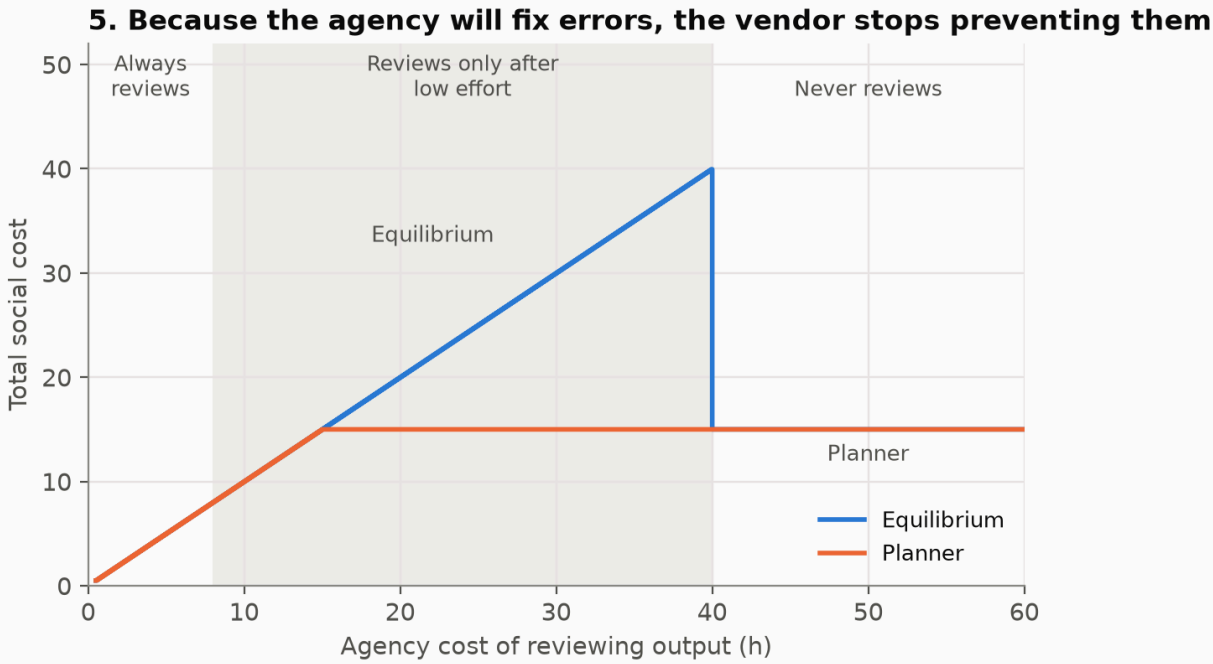
The situation. The vendor is paid per action: checks, retries, tool calls. More actions mean fewer errors. After seeing the output, the agency decides whether its staff should review and correct it, which is worthwhile only when the error rate is high (shaded band). The contract fines the vendor only for errors that **reach applicants**.

What happens.

1. If the vendor does little work, errors are high, so the agency reviews and its staff catch every error. The vendor owes nothing.
2. If the vendor does a bit more work, errors fall below the point where review is worthwhile, and the agency stops reviewing.
3. Now the remaining errors reach applicants, and the vendor pays fines on them.
4. So doing slightly more work makes the vendor worse off. With the fine in place (orange), it stops exactly at the edge of the review band and stays there over a range of prices. It does more only once the per-action pay is high enough to outweigh the fines.

Why it matters. The fine meant to protect applicants instead keeps public staff busy correcting the vendor's work. Penalize **raw** errors, measured before agency correction, or bill the vendor for the correction labor.

5. If the agency will fix bad work, the vendor stops preventing it



Model 6. In the middle regime the agency reviews only after low effort, so the vendor chooses low effort; staff labor replaces prevention.

The situation. A simpler version of the same problem. The vendor delivers high or low quality. The agency sees which, then decides whether to pay staff to review and correct the output. The horizontal axis is the agency's **cost of review (h)**.

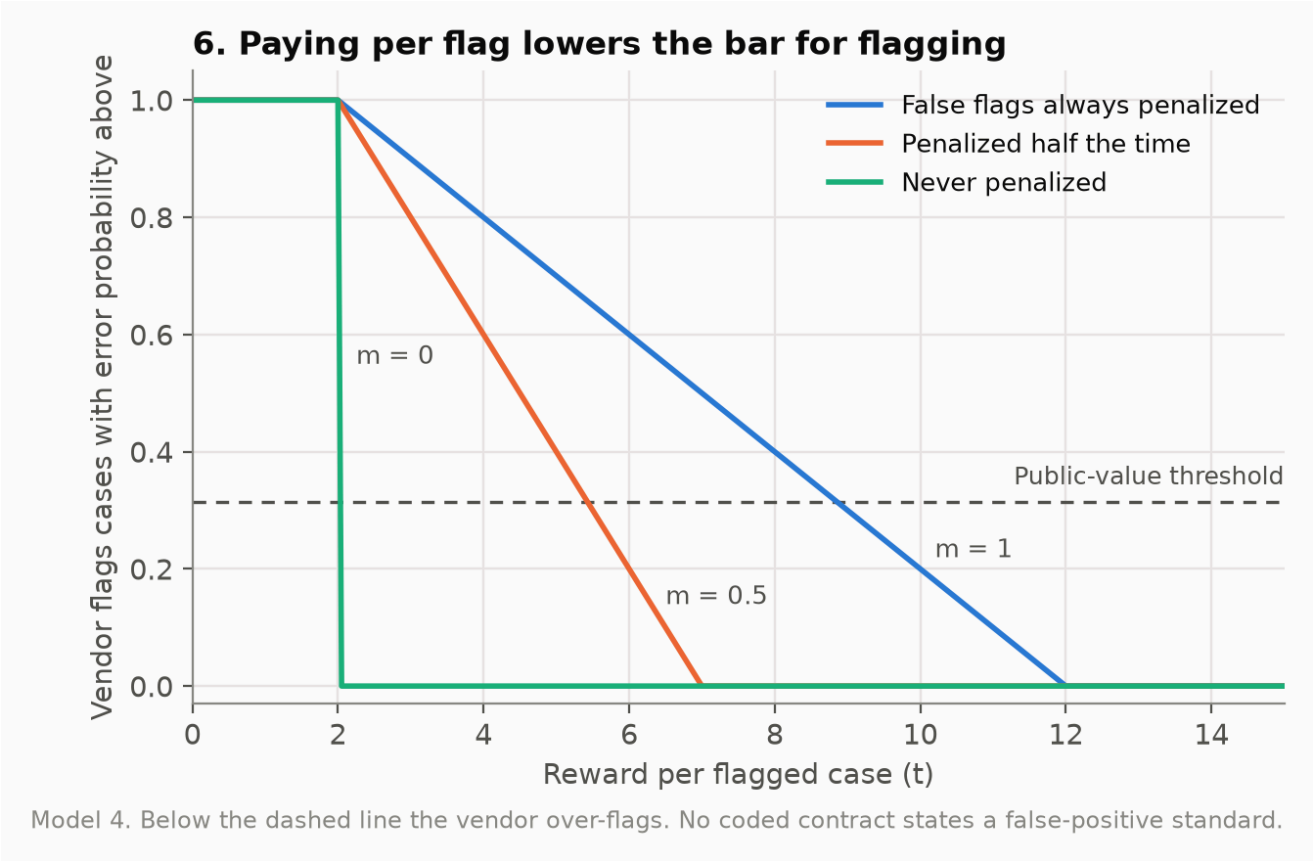
What happens. There are three regions:

- **Review is cheap (left):** the agency always reviews. Here relying on review can sensibly be the low-cost choice.
- **Review is very expensive (right):** the agency never reviews, so the vendor bears the cost of its own errors and delivers high quality.
- **In between (shaded):** the agency reviews only if quality is low. The vendor knows bad work will be rescued, like a student who knows the teacher will rewrite the essay, so it delivers low quality.

Public staff labor replaces vendor effort, and total cost (blue) rises well above what a planner would choose (orange). The gap is widest just before review becomes too expensive to bother with.

Why it matters. Systems can show few final errors while hiding large amounts of staff correction work. Measure review labor, and tie remedies to the quality of what the vendor delivers.

6. Paying per flag lowers the bar for flagging



Model 4. Below the dashed line the vendor over-flags. No coded contract states a false-positive standard.

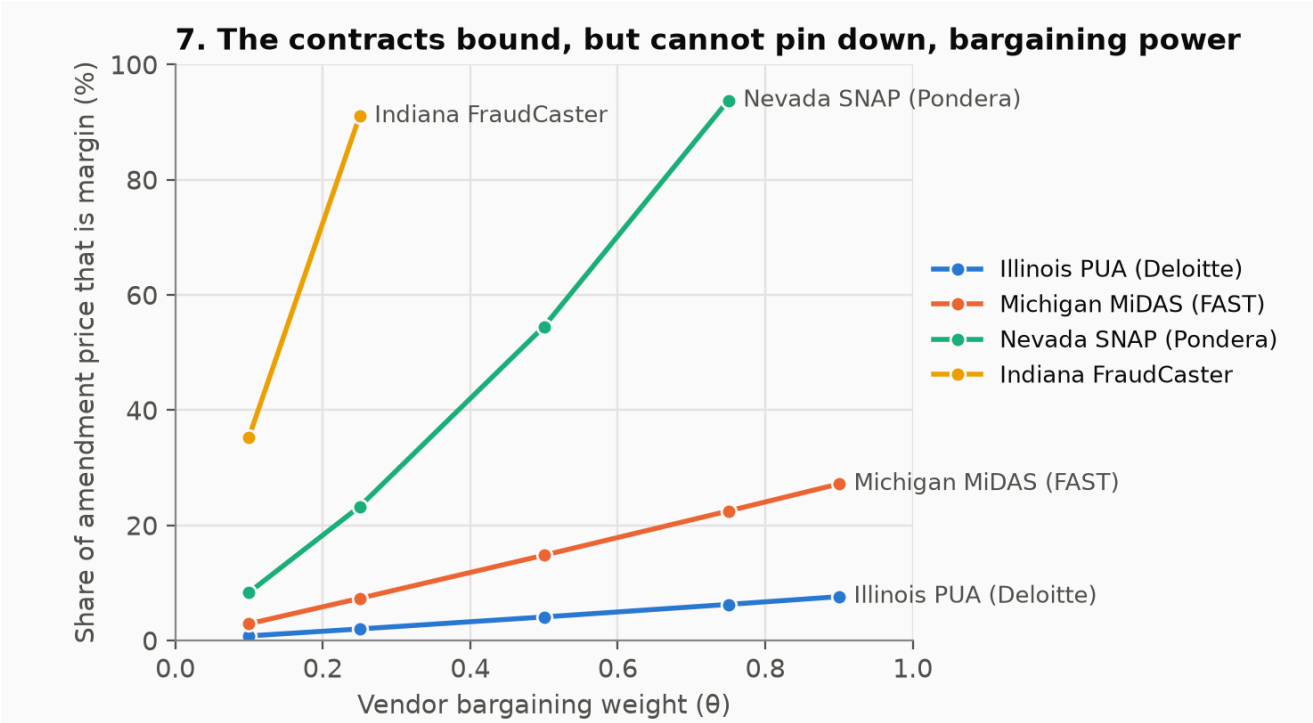
The situation. The vendor screens cases for possible payment errors. For each case it knows the probability that the case really has an error, and it decides whether to flag it. It is paid a **reward per flag (t, horizontal axis)** whether or not the flag is right. It is penalized for a false flag only if the mistake is caught (**m**, the three lines).

The vertical axis shows the **minimum probability a case needs before the vendor flags it**; lower means more flags. The dashed line is the public standard: flag only when the expected benefit of catching a real error outweighs the harm of a wrong flag, such as delays, appeals and lost benefits, plus processing cost.

What happens. The vendor flags any case where the reward beats its expected penalty for being wrong. As the reward rises it flags weaker and weaker cases, and below the dashed line it is over-flagging. If false flags are never penalized (green), it flags *every* case once the reward exceeds its processing cost.

Why it matters. Pay for findings that are validated, and set enforceable false-positive standards. None of the 19 coded state contracts states one.

7. The contract records can bound, but not pin down, vendor bargaining power



Calibration. Lines end where θ becomes inconsistent with the observed amendment. Large amendments are mostly cost at any θ.

The situation. We observe how much each real contract grew after award, but not the vendor's costs. An amendment's price is the vendor's cost plus its bargaining margin. The same dollars could mean "mostly cost, weak leverage" or "mostly margin, strong leverage".

How to read it. Pick an assumed bargaining power on the horizontal axis. The vertical axis shows what share of the amendment price would then be margin. A line stops where the assumed power would imply a margin larger than the observed amendment, which is impossible.

What it shows.

- **Large amendments are mostly cost under any assumption.** Illinois' PUA system grew 2.6× its base during the pandemic claims surge; even maximal leverage explains under 8% of that as margin.
- **Small amendments rule out strong leverage.** Indiana's $43,000 amendment is consistent only with low bargaining power, given the assumed switching costs.

Why it matters. Contract documents alone can't tell overpayment from real work. That requires cost data, invoices, or competing bids for comparable work.

Also: validation gaps compound (Model 5, not pictured)

When many agencies buy through one cooperative contract, each hopes another will pay to test the product, so shared testing is underprovided. Separately, each agency skips its own local testing when it

doesn't fully count harms to its applicants. That loss grows with the number of agencies; across the default sweep the total ranges from 1.2 to 69 cost units.

Optimal contract design (default parameters)

Problem	Optimal terms	Result
Model 1, burden weight $\omega = 0 \to 3$	Monitoring $m = 0.86 \to 1$	Agencies that weigh applicant burden monitor more
Model 2	Warranty $w = 0.77$; with a repair-price cap, $w = 0.57$ and $\mu = 0$	A price cap substitutes for costly enforcement; social cost falls from 6.33 to 4.77
Model 3	Usage price, outcome bonus, or both	Each reaches the same optimum (6.66) when costs are known
Model 4	Flag reward $t = 5.43$	Recovers the analytic aligning reward $c + mf(1 - \tau_SP)$
Model 6	Penalty $f > 60$ (64.9 reported)	Removes the review commitment problem; social cost falls from 20 to the planner's 15
Model 8	$w = 0.68$, $p = 0.63$	Joint terms beat either instrument alone
Model 9	No usage price, no penalty	Paying for activity that staff then correct is pure excess cost ($9.47 \to 6$)

Calibration to the contracts

Amendment histories (hand-verified; totals reconcile to the documents):

Contract	Base	Unplanned increases	Growth	Main triggers
Illinois IDES / Deloitte, PUA system	$9.49M	$24.34M (7 amendments)	2.57×	Pandemic claims volume, extensions
Michigan / FAST, MiDAS	$47.03M (2017 value)	$32.61M, plus $9.31M in options	0.69×	13 monthly bridge extensions and a useful-life extension; CARES Act changes
Nevada DWSS / Pondera	$1.50M	$0.39M	0.26×	Two-year extension
Indiana FSSA / FraudCaster	$0.74M	$0.04M	0.06×	Paid migration out of the legacy CODY system

Frequency. 5 of 12 contract families with observed modifications had priced unplanned work ($\lambda = 0.417$).

Coded terms across 19 state contracts:

- 0 pay for verified outcomes;
- 0 state an accuracy or false-positive standard;
- 1 contains performance liquidated damages;
- 3 contain service-level fee reductions, with caps of 5%, 10% and 20% of fees;
- 7 use unit pricing.

Cooperative purchasing is split: 2 of 7 agreements contain local acceptance-testing language. The NASPO cloud participating addenda contain none; the MMIS addendum does.

Negotiated weakening. Arizona's UI modernization memorandum records four accepted vendor deviations. Each moves a model parameter against the agency:

- source code excluded from deliverables;
- no state access to the escrow package;
- defect classification changed from state determination to mutual determination;
- the SLA at-risk cap cut from 13% to 10%.

Identified sets. The share of amendment spending attributable to bargaining margin depends on θ, which the documents cannot pin down:

Contract	Consistent θ	Margin share of amendment price
Illinois	all values tested (0.1–0.9)	0.8–7.6%
Michigan	all values tested (0.1–0.9)	2.9–27%
Nevada	up to 0.75	8–94%
Indiana	up to 0.25	35–91%

Large amendments are mostly cost under any θ. Small amendments rule out high bargaining power, given the assumed switching costs.

The binding term is the remedy. Three of four calibrated contracts have no enforceable performance penalty, so Model 1 predicts zero quality effort whatever the monitoring level. Allowing penalty exposure up to the highest cap seen in the corpus (20%) raises effort from 0 to 0.85. For MiDAS, effort rises from 0.35.

Limits

- Value, burden, cost-curvature, and governance parameters are illustrative. Design gains are reported in model units, not dollars.
- The mapping from clauses to parameters encodes explicit assumptions, not estimates. Change it in `coding_rules.json`.
- The cost-versus-margin split of amendments is not identified. Identifying it needs cost data, invoices, or competing bids.
- The documents are a FOIA/EPIC corpus, not a random sample. Some contracts are partial: the MiDAS base is its 2017 value, and Nevada's base follows Amendment 1. Several Pondera files are image scans without text and were not coded.
- Not yet modeled: structural cost estimation, partial review, alternative score distributions, realized sample paths, and other model combinations.

Part II. Designing contracts that hedge against vendor strategies

Recommendations for public agencies drawn from the procurement models (Models 1–9), the contract-design optimization, and the calibration to state contracts in this repository. Part I summarizes the workflow and results; the full models are in the repository's `game_theory/` folder.

The core principle

Make vendors pay for what the agency values: verified, correct outcomes and the full cost of achieving them, backed by remedies the agency can actually enforce. Contracts that pay for activity, flags, or nominal completion, or that rely on monitoring without consequences, invite the strategies below.

Vendor strategies and contract countermeasures

Vendor strategy	What the models show	Contract countermeasure
Bid low, earn on amendments (Model 1)	Competition pushes expected amendment margins into lower initial bids, so the award price understates lifecycle cost. Dependence still wastes resources.	Evaluate bids on **expected lifecycle cost**: base price plus priced change-order rates. Lock in hourly rates and an annual change-order budget at award, as Illinois did in its PUA contract's 7th amendment ($150–200/hr).
Build lock-in (Models 1, 8)	Dependence raises the bargained price of every later amendment and repair.	**Portability as a package:** state ownership of data *and* code, source-code escrow the agency can test and have released, transition assistance, and documentation. Open-source licensing of custom code is one way to secure the code part, not a substitute for the rest: open code can still be locked in through undocumented configuration, vendor-controlled hosting, or know-how held only by the incumbent. Arizona's UI modernization vendor won the exclusion of source code and blocked state access to the escrow package during negotiation; those are the clauses to protect.
Under-prevent defects, then bill for repairs (Models 2, 8)	Prevention collapses when uncovered defects become paid work. Weak portability makes this worse, because repairs are priced under lock-in.	**Agency-controlled defect classification**: the state decides what counts as a defect versus new scope. Make covered repairs non-billable, or **cap repair prices**. In Model 2 a price cap cut social cost from 6.33 to 4.77 and reduced the enforcement needed. Do not accept "mutually determine" language.
Shirk where monitoring has no consequence (Model 1, calibration)	With no enforceable penalty, quality effort is zero at any monitoring level. Three of the four calibrated contracts were in this position.	**Remedies first, monitoring second.** Service-level fee reductions with meaningful caps: the corpus shows 5–20% of fees, and a 20% cap raised modeled effort from 0 to 0.85. Then fund independent verification and validation (IV&V) so failures are detected.
Inflate billable activity (Models 3, 9)	Per-action pay rewards extra checks, retries, and tool calls, especially when staff correct the errors anyway.	Pay for **verified correct completion**. If you use usage pricing, set the rate knowing costs and burdens, and track **cost per correct case**, not call counts.

Vendor strategy	What the models show	Contract countermeasure
Let agency staff absorb errors (Models 6, 9)	A penalty only on errors that reach applicants is neutralized by the agency's own review. Vendors can hold quality just low enough that staff keep correcting.	Tie remedies to **raw error rates** measured before agency correction, or **charge the vendor for correction labor**. Record staff review time as a contract cost.
Over-flag to hit a proxy (Model 4)	A per-flag reward with no false-positive penalty means flag everything. None of the 19 coded state contracts states an accuracy or false-positive standard.	Pay for **validated** findings, include an enforceable false-positive standard, and measure appeal-adjusted outcomes on representative samples, not just flag counts.
Swap in a cheaper configuration after acceptance (Model 7)	Deterrence requires audit probability × sanction ≥ the vendor's savings. Termination threats count only if credible.	**Change-notification and revalidation duties**, audit rights with evidence access, and remedies that do not depend on termination, since lock-in makes termination non-credible.
Free-ride on a cooperative award (Model 5)	Shared tests are underprovided, and local tests are skipped when agencies discount local harms. The NASPO cloud participating addenda contain no local testing language.	**Assign and fund** common validation explicitly, and require **local acceptance testing** in participating addenda. Treat purchasing eligibility as distinct from validated deployment.

Design principles

1. **Instruments interact, so design them together.** Warranty enforcement and portability are complements (Model 8): portability raises prevention, and enforcement is weaker without it. Penalties and review interact (Model 9). Strengthening one term while its partner stays weak can accomplish nothing.
2. **Remedies bind before detection.** Spending on monitoring is wasted without an enforceable, adequately capped consequence.
3. **Measure the right outcome.** Use raw correctness, verified completion, cost per correct case, and appeal-adjusted accuracy, not activity, flags, or errors that survive review.
4. **Protect key terms during negotiation.** Arizona's memorandum of negotiations shows value leaking through four accepted vendor deviations: source code excluded, escrow access blocked, defect classification shared, and the SLA at-risk cap cut from 13% to 10%. Treat such terms as non-negotiable, or price them explicitly.
5. **Keep a credible exit.** Portability, transition plans, and competitive re-procurement timelines prevent extension chains like MiDAS's 13 monthly bridge extensions in 2021–22.

Contract checklist

At solicitation and award

- ☐ Bids are scored on expected lifecycle cost, including priced change-order rates and an annual change-order budget.

- [ ] The state owns data, deliverables, and custom code; escrow is testable and has defined release events.
- [ ] Transition assistance, documentation, and knowledge transfer are deliverables with acceptance criteria.

Quality and remedies

- [ ] The agency alone classifies defects versus new scope; covered repairs are non-billable or price-capped.
- [ ] Service-level fee reductions have meaningful caps and are actually assessed.
- [ ] Remedies attach to raw error rates, not only errors that reach applicants.
- [ ] Accuracy and false-positive standards are stated and measured on representative samples.
- [ ] IV&V or equivalent independent testing is funded.

Payment structure

- [ ] Payment is tied to verified correct completion, not activity volume or flag counts.
- [ ] Agency correction and review labor is measured, and charged back where attributable.

After award

- [ ] Material changes to models, data sources, or subcontractors require notification and revalidation.
- [ ] Audit rights include access to logs and evidence.
- [ ] Remedies exist that do not depend on termination.
- [ ] Re-procurement starts early enough to avoid bridge extensions.

Cooperative purchasing

- [ ] Common validation is assigned and funded.
- [ ] Participating addenda require local acceptance testing before deployment.

Limits of contractual safeguards

Contracts can align incentives on outcomes that are **observable and verifiable**, but only while the agency can still exit. Beyond that, safeguards hit predictable limits.

Limits that come from the models

Limit	Why it happens	Where it shows up
Safeguards aren't free	With competitive bidding, vendors price risk into their bids, so the agency pays for every penalty and warranty. Penalties above the real harm cause over-investment and may thin the bidder pool.	Model 1 bid offset; Model 2 overinvestment
A remedy is only as good as the evidence behind it	Effort responds to *chance of detection* × *penalty*. If errors are hard to observe or their classification is disputed, nominal clauses do nothing.	Model 1 (m·F); Model 2 (w is enforced coverage, not contract wording)
Lock-in undermines enforcement	Termination and re-procurement threats are credible only if the agency can actually leave. The more dependent it is, the weaker every exit-based remedy.	Model 7: with larger cost savings or weaker detection, even auditing every time cannot deter substitution

Limit	Why it happens	Where it shows up
Contracts get renegotiated	Amendments are bargained under lock-in, so protections can be traded away mid-contract or before signature.	Model 1 bargaining
The agency can't bind its own future behavior	Once bad output arrives, fixing it is the agency's best move. The vendor anticipates this, so remedies that apply only after the agency fails to fix things are neutralized.	Models 6 and 9: the vendor holds work at the review cutoff
Every measurable target can be gamed	Pay for flags and you get flags; penalize only errors that reach applicants and the vendor manages around review.	Models 4 and 9
One term alone often does nothing	Safeguards work in pairs: warranty with portability, penalty with detection, reward with false-flag enforcement.	Model 8; Model 4 (the optimal per-flag reward falls from 8.9 to 2.7 when false flags are rarely caught)
Unforeseen change can't be priced in advance	Pandemic programs, new federal guidance, and caseload spikes create genuinely new work. Contracts can only set *how* it is priced, through rate cards and change-order budgets.	Illinois PUA amendments; MiDAS CARES Act changes
One buyer can't fix a collective problem	No single agency's contract funds testing whose benefits are shared across buyers.	Model 5

What the contract data add

- **Remedies are rare and capped.** 3 of 19 state contracts have service-level fee reductions, capped at 5–20% of fees. Only 1 has performance liquidated damages. None pays for outcomes or states an accuracy standard.
- **Protections erode in negotiation.** Arizona accepted four vendor deviations, each weakening a safeguard.
- **Continuity overrides competition.** MiDAS ran through 13 monthly bridge extensions "while negotiations continue". The exit threat that safeguards rely on wasn't available.
- **The counterparty can change.** Pondera contracts were reassigned to West Publishing (Thomson Reuters) mid-term after an acquisition, and Kansas bought through a reseller (SHI). Both blur who is responsible for remedies.
- **Burden falls outside the contract.** Errors that hit applicants appear in no contract metric unless the agency writes them in.

Safeguards therefore need non-contractual supports:

- in-house technical capacity;
- independent testing;
- re-procurement planning that starts early;
- representative audits of outcomes for applicants.

Does the optimal design differ for AI?

What the data can and cannot say

The corpus is mostly **predictive risk scoring and legacy systems**, not generative or agentic AI:

- **Predictive scoring:** Pondera, FraudCaster, COMPAS, Vant4ge, and OnPoint.
- **Large, mostly rules-based builds:** MiDAS, the Deloitte UI systems, and HEAplus.

The data therefore cannot compare AI with non-AI contracts, and they say nothing about generative or agentic systems. They do support one comparison: predictive-analytics subscriptions versus system builds.

Coded safeguard	Predictive analytics (11)	System builds (7)
Any performance remedy (liquidated damages or service-level penalties)	**0**	3
Transition assistance	2	6
Source-code escrow	1	3
Independent verification and validation	0	2
Unit pricing	4	3
Accuracy or false-positive standard	0	0

The scoring products, the most AI-like contracts in the corpus, carry the *fewest* safeguards. The comparison is confounded: they are small vendor-standard subscriptions, while the builds are custom projects worth $50 million or more.

What the theory says

The core principle doesn't change: pay for verified outcomes, back them with enforceable remedies, and keep a credible exit. What changes is the parameters, so the binding term and its required strength shift.

The table below re-runs the design optimizer under parameter shifts that plausibly distinguish the technologies. The shifts are illustrative, not estimated.

Technology	What plausibly differs	What the models say to emphasize
Legacy builds	Lock-in and defect-driven amendments dominate; errors are relatively testable	Warranty enforcement, portability, and rate cards (Models 1, 2, 8). Optimal portability rises with dependence leverage, from 0.19 to 0.82 as leverage goes from 1 to 4.
Predictive scoring	Payment tied to flags; false positives are hard to observe and harm applicants	Pay for validated findings, and set the reward together with false-flag enforcement. When false flags are rarely caught, the optimal per-flag reward falls by about two-thirds (Model 4).
Generative AI	Output errors are pervasive, verifying them is costly, and providers swap models often	Remedies on raw error rates, measured staff review, and change notification (Models 6, 7, 9). Raising the vendor's savings from a cheaper model from 4 to 8, or cutting detection from 0.5 to 0.2, makes deterrence impossible even with certain auditing. Notification that raises detection to 0.8, plus a larger sanction, restores it.
Agentic AI	The vendor's orchestration controls how much billable work happens, and actions can cause harm	Avoid usage pricing, cap and log actions, attach liability to actions, and require recovery duties. The optimum stays at no per-action price, but a bad price becomes far costlier when the vendor controls more of the workflow: at the same price, welfare loss rises from 7.4 to 19.3 (Model 9).

Where theory is ambiguous: does AI raise or lower lock-in?

- **Raises it:** dependence on retrieval systems, prompts, evaluation sets, and workflow state.
- **Lowers it:** interchangeable foundation models and cheaper development.

The models cannot settle this, and it determines whether portability terms matter more or less for AI. It is the main empirical question for a follow-up.

What a real comparison would need

Generative and agentic contracts with their amendments, usage invoices, evaluation reports, and permission settings, classified by actual capability and deployment date rather than by signing year.

Caveats

- **Magnitudes are illustrative.** Figures such as "a 20% cap raises effort to 0.85" depend on uncalibrated value, cost, and burden parameters. The direction of each recommendation is robust within the models; the levels are not.
- **Stronger terms have costs.** Vendors price risk into bids, and penalties above the true social loss produce overinvestment (Model 2). Very demanding terms may also reduce the number of bidders, which these models do not capture.
- **Evidence gaps.** The contract records cannot separate amendment costs from bargaining margins, and the corpus is a FOIA/EPIC sample rather than a random one. Testing these recommendations would require cost data, invoices or competing bids for comparable work, and records of how often agencies actually enforce remedies.